

Chapter 8 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 8 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter8_Knowledge`. The cited knowledge IDs refer to entries in `chapter8_knowledge_pieces.jsonl`.

Exercise 8.1.1

Question. Use Exercise 7.5.4 to conclude that

$$\mathbb{E}T_{U,V}^2 \leq 4\mathbb{E}X^4.$$

Knowledge used. `durrett_ch8_skorokhod_embedding_mean_zero`;
`durrett_ch8_embedded_random_walk`.

Solution. Recall the notation from the proof of Skorokhod's representation theorem. The mean-zero law of X is written as a mixture of two-point mean-zero laws $\mu_{u,v}$, where $u < 0 < v$ and

$$\mu_{u,v}(\{u\}) = \frac{v}{v-u}, \quad \mu_{u,v}(\{v\}) = \frac{-u}{v-u}.$$

Given $(U, V) = (u, v)$, the stopping time is

$$T_{u,v} = \inf\{t : B_t \notin (u, v)\},$$

so $B_{T_{u,v}}$ has law $\mu_{u,v}$.

Exercise 7.5.4 says that for Brownian motion exiting an interval (a, b) with $a < 0 < b$,

$$\mathbb{E}T_{a,b}^2 \leq 4\mathbb{E}B_{T_{a,b}}^4.$$

Applying this conditional on $(U, V) = (u, v)$ gives

$$\mathbb{E}(T_{U,V}^2 \mid U, V) \leq 4\mathbb{E}(B_{T_{U,V}}^4 \mid U, V) = 4 \int x^4 \mu_{U,V}(dx).$$

Now take expectations over the mixing distribution of (U, V) . Since the mixture of the laws $\mu_{U,V}$ is exactly the original law of X ,

$$\mathbb{E}T_{U,V}^2 \leq 4\mathbb{E} \int x^4 \mu_{U,V}(dx) = 4\mathbb{E}X^4.$$

If $\mathbb{E}X^4 = \infty$, the inequality is still true in the extended sense; otherwise it gives the desired finite second-moment bound for the embedding time.

Exercise 8.1.2

Question. Suppose S_n is one-dimensional simple random walk and let

$$R_n = 1 + \max_{m \leq n} S_m - \min_{m \leq n} S_m$$

be the number of points visited by time n . Show that $R_n/\sqrt{n}$ converges in distribution to a limit.

Knowledge used. [durrett_ch8_donsker_invariance_principle](#);
[durrett_ch8_continuous_functional_transfer](#); [durrett_ch8_maximum_functional](#).

Solution. For one-dimensional nearest-neighbor simple random walk, every integer between the running minimum and running maximum has been visited. Hence the number of visited points by time n is exactly

$$R_n = 1 + \max_{0 \leq m \leq n} S_m - \min_{0 \leq m \leq n} S_m.$$

Let $W_n(t) = S(\lfloor nt \rfloor)/\sqrt{n}$ with linear interpolation between the mesh points. Since the increments of simple random walk have mean 0 and variance 1, Donsker's theorem gives

$$W_n \Rightarrow B \quad \text{in } C[0, 1],$$

where B is standard Brownian motion.

Define the range functional

$$\psi(\omega) = \sup_{0 \leq t \leq 1} \omega(t) - \inf_{0 \leq t \leq 1} \omega(t), \quad \omega \in C[0, 1].$$

This functional is continuous under the uniform norm. Indeed, if $\|\omega - \eta\|_\infty \leq \varepsilon$, then

$$|\sup_t \omega(t) - \sup_t \eta(t)| \leq \varepsilon, \quad |\inf_t \omega(t) - \inf_t \eta(t)| \leq \varepsilon,$$

and therefore $|\psi(\omega) - \psi(\eta)| \leq 2\varepsilon$. The continuous mapping theorem now implies

$$\max_{0 \leq m \leq n} \frac{S_m}{\sqrt{n}} - \min_{0 \leq m \leq n} \frac{S_m}{\sqrt{n}} = \psi(W_n) \Rightarrow \psi(B) = \sup_{0 \leq t \leq 1} B_t - \inf_{0 \leq t \leq 1} B_t.$$

Finally, $1/\sqrt{n} \rightarrow 0$, so Slutsky's theorem gives

$$\frac{R_n}{\sqrt{n}} = \frac{1}{\sqrt{n}} + \max_{0 \leq m \leq n} \frac{S_m}{\sqrt{n}} - \min_{0 \leq m \leq n} \frac{S_m}{\sqrt{n}} \Rightarrow \sup_{0 \leq t \leq 1} B_t - \inf_{0 \leq t \leq 1} B_t.$$

Thus $R_n/\sqrt{n}$ converges in distribution. The limiting random variable is the range of Brownian motion on $[0, 1]$.

Section 8.2 Exercises

Question. The local textbook PDF has no separately numbered exercises in Section 8.2. Section 8.2, “CLTs for Martingales,” begins immediately after Exercise 8.1.2 and then proceeds through Theorems 8.2.1–8.2.8 before Section 8.3 begins.

Knowledge used. [durrett_ch8_martingale_sk_embedding](#);
[durrett_ch8_martingale_convergence_variance_sum](#); [durrett_ch8_martingale_fclt_random_clock](#);
[durrett_ch8_martingale_lindeberg_feller](#); [durrett_ch8_martingale_clt_final_form](#).

Solution. There are therefore no Section 8.2 exercise statements to solve from this source file. The main usable knowledge from the section is the martingale CLT toolkit:

- (i) square-integrable martingales can be embedded in Brownian motion at stopping times;
- (ii) the Brownian clock is controlled by conditional variances;
- (iii) a martingale functional CLT follows when the conditional variance process converges to deterministic time and large jumps are negligible;
- (iv) the Lindeberg-Feller martingale theorem is obtained by truncation;
- (v) the single-sequence martingale CLT follows by converting the sequence into a triangular array.

These are recorded above as the relevant Viki knowledge IDs, but the textbook does not give an Exercise 8.2.1 or any other Section 8.2 exercise to solve.

Exercise 8.3.1

Question. Consider the special case of Example 8.3.3 in which $\xi_i, i \in \mathbb{Z}$, are i.i.d. and take values H and T with equal probability. Let

$$\eta_n = f(\xi_n, \xi_{n+1}), \quad f(H, T) = 1, \quad f(i, j) = 0 \text{ otherwise.}$$

Then $S_n = \eta_1 + \cdots + \eta_n$ counts the number of head runs in $\xi_1, \dots, \xi_{n+1}$. Apply Theorem 8.3.2 with $\mathcal{F}_n = \sigma(\xi_m : m \leq n+1)$ to show that there are constants μ and σ so that

$$\frac{S_n - n\mu}{\sigma n^{1/2}} \Rightarrow \chi.$$

What is the random variable Y_0 constructed in the proof of Theorem 8.3.2 in this case?

Knowledge used. `durrett_ch8_stationary_coboundary_clt`;
`durrett_ch8_projective_stationary_clt`; `durrett_ch8_m_dependent_clt`;
`durrett_ch8_martingale_clt_final_form`.

Solution. Let

$$X_n = \eta_n - \mu, \quad \mu = \mathbb{E}\eta_n.$$

Since $\eta_n = 1_{\{\xi_n=H, \xi_{n+1}=T\}}$,

$$\mu = \mathbb{P}(\xi_n = H, \xi_{n+1} = T) = \frac{1}{4}.$$

The sequence (X_n) is stationary and ergodic because it is a shift-factor of the i.i.d. sequence (ξ_n) . Also X_0 depends only on ξ_0, ξ_1 . With $\mathcal{F}_n = \sigma(\xi_m : m \leq n+1)$, for $k \geq 2$ the σ -field $\mathcal{F}_{-k}$ is generated by variables ξ_m with $m \leq -k+1 \leq -1$, and hence is independent of X_0 . Therefore

$$\mathbb{E}(X_0 \mid \mathcal{F}_{-k}) = 0, \quad k \geq 2.$$

Thus

$$\sum_{k \geq 1} \|\mathbb{E}(X_0 \mid \mathcal{F}_{-k})\|_2 < \infty,$$

so Theorem 8.3.2 applies.

It remains to compute the constants. We already have $\mu = 1/4$. Since (η_n) is 1-dependent, only the lag-one covariance contributes:

$$\sigma^2 = \text{Var}(\eta_0) + 2 \text{Cov}(\eta_0, \eta_1).$$

Now

$$\text{Var}(\eta_0) = \frac{1}{4} \left(1 - \frac{1}{4}\right) = \frac{3}{16}.$$

The events $\{\eta_0 = 1\}$ and $\{\eta_1 = 1\}$ cannot occur together, because they would require $\xi_1 = T$ and $\xi_1 = H$ simultaneously. Hence $\mathbb{E}(\eta_0\eta_1) = 0$, and

$$\text{Cov}(\eta_0, \eta_1) = 0 - \frac{1}{4} \cdot \frac{1}{4} = -\frac{1}{16}.$$

Therefore

$$\sigma^2 = \frac{3}{16} + 2 \left(-\frac{1}{16}\right) = \frac{1}{16}, \quad \sigma = \frac{1}{4}.$$

Theorem 8.3.2 gives

$$\frac{S_n - n/4}{(1/4)\sqrt{n}} \Rightarrow \chi,$$

where χ has the standard normal distribution.

Finally we identify the Y_0 from the proof of Theorem 8.3.2. Write $I_j = 1_{\{\xi_j=H\}}$, so $\eta_j = I_j(1 - I_{j+1})$ and $X_j = \eta_j - 1/4$. The proof defines

$$Y_0 = \sum_{j=0}^{\infty} \{\mathbb{E}(X_j | \mathcal{F}_0) - \mathbb{E}(X_j | \mathcal{F}_{-1})\}.$$

All terms with $j \geq 2$ vanish, because X_j is independent of $\mathcal{F}_0$. For $j = 0$,

$$\mathbb{E}(X_0 | \mathcal{F}_0) = X_0, \quad \mathbb{E}(X_0 | \mathcal{F}_{-1}) = \frac{1}{2}I_0 - \frac{1}{4}.$$

For $j = 1$,

$$\mathbb{E}(X_1 | \mathcal{F}_0) = \frac{1}{2}I_1 - \frac{1}{4}, \quad \mathbb{E}(X_1 | \mathcal{F}_{-1}) = 0.$$

Thus

$$Y_0 = \left(I_0(1 - I_1) - \frac{1}{4}\right) - \left(\frac{1}{2}I_0 - \frac{1}{4}\right) + \left(\frac{1}{2}I_1 - \frac{1}{4}\right),$$

or equivalently

$$Y_0 = \frac{1}{2}I_0 + \frac{1}{2}I_1 - I_0I_1 - \frac{1}{4}.$$

This takes the value $1/4$ when $\xi_0 \neq \xi_1$ and $-1/4$ when $\xi_0 = \xi_1$. In particular $\mathbb{E}Y_0^2 = 1/16$, matching the limiting variance computed above.

Exercise 8.4.1

Question. Use Exercise 7.4.8 and the reasoning that led to (8.4.11) to conclude

$$\mathbb{P} \left(\max_{0 \leq t \leq 1} B_t^0 > b \right) = \exp(-2b^2),$$

where B^0 is Brownian bridge.

Knowledge used. `durrett_ch8_brownian_bridge_conditioning`;
`durrett_ch8_kolmogorov_distribution_method`.

Solution. Think of Brownian bridge as Brownian motion conditioned on $B_1 = 0$. Let $T_b = \inf\{t : B_t = b\}$. For Brownian motion starting at 0,

$$\left\{ \max_{0 \leq t \leq 1} B_t > b \right\} = \{T_b < 1\}.$$

Exercise 7.4.8 gives the reflected density

$$\mathbb{P}_0(T_b < 1, B_1 \in dx) = p_1(0, 2b - x) dx, \quad x < b.$$

Conditioning on $B_1 = 0$ therefore gives

$$\mathbb{P} \left(\max_{0 \leq t \leq 1} B_t^0 > b \right) = \frac{p_1(0, 2b)}{p_1(0, 0)}.$$

Since $p_1(0, x) = (2\pi)^{-1/2} \exp(-x^2/2)$,

$$\frac{p_1(0, 2b)}{p_1(0, 0)} = \exp \left(-\frac{(2b)^2}{2} \right) = \exp(-2b^2).$$

Exercise 8.4.2

Question. Let B_t be a Brownian motion starting at 0. Show that

$$X_t = (1 - t)B \left(\frac{t}{1 - t} \right), \quad 0 \leq t < 1,$$

is a Brownian bridge.

Knowledge used. [durrett_ch8_brownian_bridge_conditioning](#).

Solution. The process X is Gaussian and has mean zero, since it is obtained by deterministic linear transformations of Brownian motion. For $0 \leq s \leq t < 1$,

$$\mathbb{E}(X_s X_t) = (1 - s)(1 - t) \mathbb{E} \left[B \left(\frac{s}{1 - s} \right) B \left(\frac{t}{1 - t} \right) \right].$$

Because $u \mapsto u/(1 - u)$ is increasing,

$$\mathbb{E} \left[B \left(\frac{s}{1 - s} \right) B \left(\frac{t}{1 - t} \right) \right] = \frac{s}{1 - s}.$$

Thus

$$\mathbb{E}(X_s X_t) = (1 - s)(1 - t) \frac{s}{1 - s} = s(1 - t),$$

which is exactly the covariance of Brownian bridge $B_t^0 = B_t - tB_1$.

Finally, as $t \uparrow 1$ let $u = t/(1 - t)$. Then

$$X_t = \frac{B_u}{1 + u}.$$

Brownian motion is sublinear at infinity, so $B_u/(1 + u) \rightarrow 0$ almost surely. Setting $X_1 = 0$, the process has continuous paths on $[0, 1]$. Since it is a centered Gaussian process with Brownian-bridge covariance, X is a Brownian bridge.

Exercise 8.4.3

Question. Show that Brownian bridge is a Markov process by checking that if $s_1 < \dots < s_n < s < t < 1$, then

$$\mathbb{P}(B^0(t) = y \mid B^0(s) = x, B^0(s_n) = x_n, \dots, B^0(s_1) = x_1) = \mathbb{P}(B^0(t) = y \mid B^0(s) = x).$$

Knowledge used. [durrett_ch8_brownian_bridge_conditioning](#);
[durrett_ch8_kolmogorov_distribution_method](#).

Solution. Use the pinned Brownian transition-density formula. If $0 = t_0 < s_1 < \dots < s_n < s < t < 1$ and $x_0 = 0$, the joint density of

$$B^0(s_1), \dots, B^0(s_n), B^0(s), B^0(t)$$

at $x_1, \dots, x_n, x, y$ is proportional to

$$\left\{ \prod_{j=1}^n p_{s_j - s_{j-1}}(x_{j-1}, x_j) \right\} p_{s-s_n}(x_n, x) p_{t-s}(x, y) p_{1-t}(y, 0).$$

The joint density of $B^0(s_1), \dots, B^0(s_n), B^0(s)$ at $x_1, \dots, x_n, x$ is proportional to

$$\left\{ \prod_{j=1}^n p_{s_j - s_{j-1}}(x_{j-1}, x_j) \right\} p_{s-s_n}(x_n, x) p_{1-s}(x, 0).$$

Dividing the two densities, all factors involving the earlier times $s_1, \dots, s_n$ cancel. Hence the conditional density of $B^0(t)$ given the displayed past is

$$\frac{p_{t-s}(x, y) p_{1-t}(y, 0)}{p_{1-s}(x, 0)},$$

which depends only on the present value $B^0(s) = x$. This proves the Markov property.

Exercise 8.4.4

Question. Let B_t^0 be Brownian bridge. Show that

$$X_t = \frac{B_t^0}{1-t}, \quad 0 \leq t < 1,$$

is a martingale but is not L^2 bounded.

Knowledge used. [durrett_ch8_brownian_bridge_conditioning](#).

Solution. Let $\mathcal{F}_t^0 = \sigma(B_u^0 : 0 \leq u \leq t)$ be the natural filtration of the bridge. From the Brownian-bridge transition law, for $0 \leq s < t < 1$,

$$\mathbb{E}(B_t^0 \mid \mathcal{F}_s^0) = \mathbb{E}(B_t^0 \mid B_s^0) = \frac{1-t}{1-s} B_s^0.$$

Therefore

$$\mathbb{E}(X_t \mid \mathcal{F}_s^0) = \frac{1}{1-t} \mathbb{E}(B_t^0 \mid \mathcal{F}_s^0) = \frac{B_s^0}{1-s} = X_s.$$

Thus $(X_t)_{0 \leq t < 1}$ is a martingale.

It is not L^2 bounded. Since Brownian bridge has

$$\text{Var}(B_t^0) = t(1-t),$$

we have

$$\mathbb{E}X_t^2 = \frac{\mathbb{E}(B_t^0)^2}{(1-t)^2} = \frac{t(1-t)}{(1-t)^2} = \frac{t}{1-t}.$$

As $t \uparrow 1$, this tends to infinity. Hence

$$\sup_{0 \leq t < 1} \mathbb{E}X_t^2 = \infty,$$

so the martingale is not bounded in L^2 .

Exercise 8.5.1

Question. Show that if $\mathbb{E}|X_i|^\alpha = \infty$ for some $\alpha < 2$, then

$$\limsup_{n \rightarrow \infty} \frac{|X_n|}{n^{1/\alpha}} = \infty \quad \text{a.s.}$$

so the law of the iterated logarithm fails.

Knowledge used. [durrett_ch8_iid_lil](#); [durrett_ch8_brownian_lil](#).

Solution. Let $Y = |X_1|^\alpha$. Since $\mathbb{E}Y = \infty$ and the tail function $t \mapsto \mathbb{P}(Y > t)$ is nonincreasing,

$$\sum_{n=1}^{\infty} \mathbb{P}(Y > cn) = \infty \quad \text{for every } c > 0.$$

Equivalently, for every $M > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > Mn^{1/\alpha}) = \sum_{n=1}^{\infty} \mathbb{P}(Y > M^\alpha n) = \infty.$$

The events $\{|X_n| > Mn^{1/\alpha}\}$ are independent, so the second Borel-Cantelli lemma gives

$$|X_n| > Mn^{1/\alpha} \quad \text{infinitely often a.s.}$$

Since M is arbitrary, applying this to $M = 1, 2, 3, \dots$ gives

$$\limsup_{n \rightarrow \infty} \frac{|X_n|}{n^{1/\alpha}} = \infty \quad \text{a.s.}$$

This rules out the LIL normalization. Indeed, if an increment is very large, then at least one of the two adjacent partial sums is large:

$$|X_n| = |S_n - S_{n-1}| \leq |S_n| + |S_{n-1}|,$$

so

$$\max(|S_n|, |S_{n-1}|) \geq \frac{|X_n|}{2}.$$

Since $\alpha < 2$,

$$\frac{n^{1/\alpha}}{\sqrt{n \log \log n}} = \frac{n^{1/\alpha-1/2}}{\sqrt{\log \log n}} \rightarrow \infty.$$

Along the infinitely many large-jump indices above, the LIL scale $\sqrt{2n \log \log n}$ is therefore too small to keep the partial sums bounded. Hence the finite-variance LIL conclusion cannot hold.

Exercise 8.5.2

Question. Give a direct proof that, under the hypotheses of Theorem 8.5.3, the limit set of

$$\left\{ \frac{S_n}{(2n \log \log n)^{1/2}} : n \geq 3 \right\}$$

is $[-1, 1]$.

Knowledge used. [durrett_ch8_iid_lil](#); [durrett_ch8_strassen_invariance_principle](#).

Solution. Theorem 8.5.3 says that the limit set in $C[0, 1]$ of

$$Z_n(t) = \frac{S(nt)}{(2n \log \log n)^{1/2}}$$

is

$$K = \left\{ f : f(x) = \int_0^x g(y) dy, \int_0^1 g(y)^2 dy \leq 1 \right\}.$$

The map $e_1 : C[0, 1] \rightarrow \mathbb{R}$ given by $e_1(f) = f(1)$ is continuous. Therefore the scalar subsequential limits of

$$\frac{S_n}{(2n \log \log n)^{1/2}} = Z_n(1)$$

are exactly the values $f(1)$ with $f \in K$.

First let $f \in K$. Then $f(1) = \int_0^1 g(y) dy$ for some g with $\int_0^1 g^2 \leq 1$. By Cauchy's inequality,

$$|f(1)| = \left| \int_0^1 g(y) dy \right| \leq \left(\int_0^1 g(y)^2 dy \right)^{1/2} \leq 1.$$

Thus every scalar limit point lies in $[-1, 1]$.

Conversely, fix $a \in [-1, 1]$ and set $f_a(t) = at$. Then $f_a(t) = \int_0^t a dy$ and

$$\int_0^1 a^2 dy = a^2 \leq 1,$$

so $f_a \in K$. By Theorem 8.5.3, there is a subsequence n_k such that $Z_{n_k} \rightarrow f_a$ uniformly on $[0, 1]$. Evaluating at 1 gives

$$\frac{S_{n_k}}{(2n_k \log \log n_k)^{1/2}} = Z_{n_k}(1) \rightarrow f_a(1) = a.$$

Hence every $a \in [-1, 1]$ is a limit point, and the scalar limit set is exactly $[-1, 1]$.